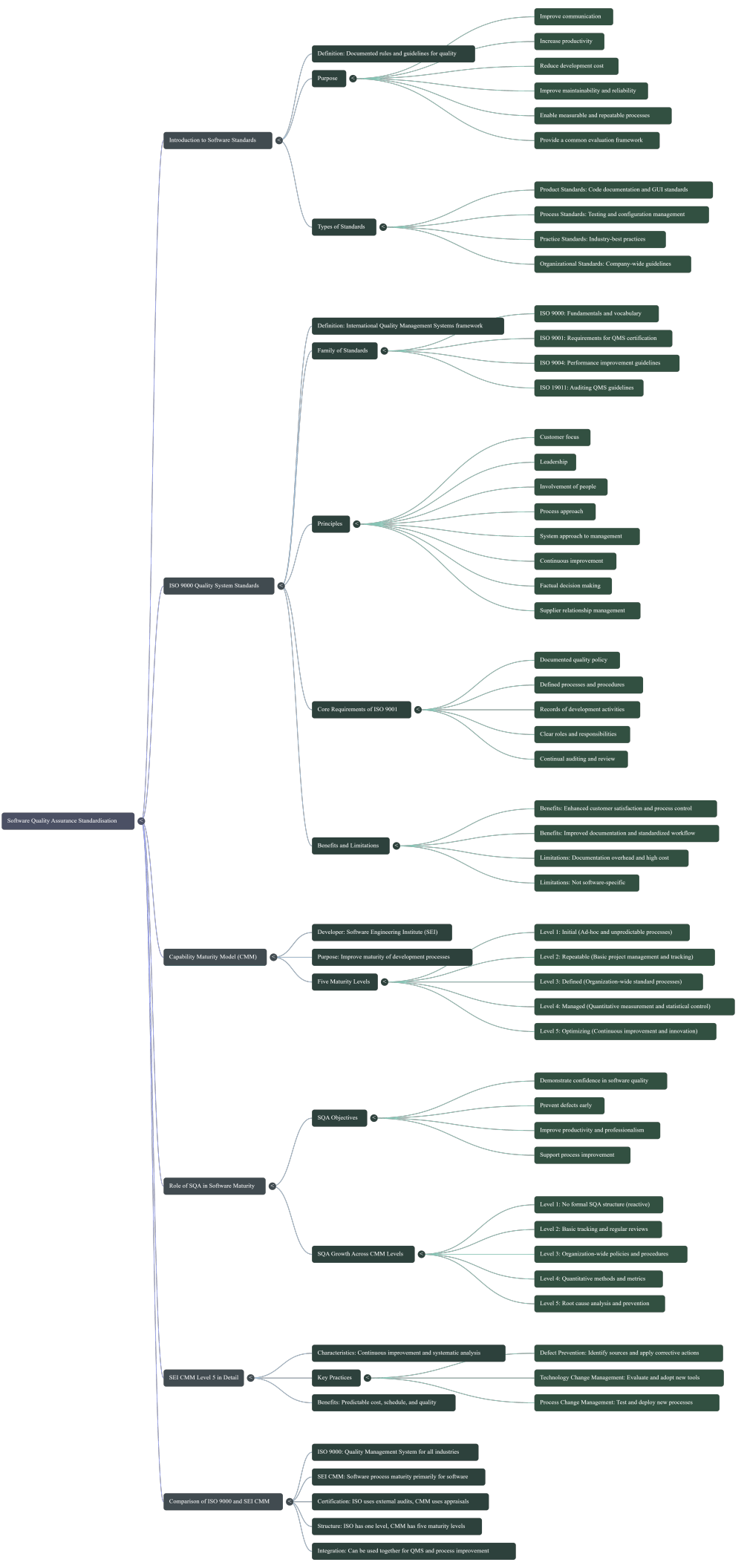

Software Quality Assurance Standardisation
Introduction to Software Standards
Definition: Documented rules and guidelines for quality
Purpose
Improve communication
Increase productivity
Reduce development cost
Improve maintainability and reliability
Enable measurable and repeatable processes
Provide a common evaluation framework
Types of Standards
Product Standards: Code documentation and GUI standards
Process Standards: Testing and configuration management
Practice Standards: Industry-best practices
Organizational Standards: Company-wide guidelines
ISO 9000 Quality System Standards
Definition: International Quality Management Systems framework
Family of Standards
ISO 9000: Fundamentals and vocabulary
ISO 9001: Requirements for QMS certification
ISO 9004: Performance improvement guidelines
ISO 19011: Auditing QMS guidelines
Principles
Customer focus
Leadership
Involvement of people
Process approach
System approach to management
Continuous improvement
Factual decision making
Supplier relationship management
Core Requirements of ISO 9001
Documented quality policy
Defined processes and procedures
Records of development activities
Clear roles and responsibilities
Continual auditing and review
Benefits and Limitations
Benefits: Enhanced customer satisfaction and process control
Benefits: Improved documentation and standardized workflow
Limitations: Documentation overhead and high cost
Limitations: Not software-specific
Capability Maturity Model (CMM)
Developer: Software Engineering Institute (SEI)
Purpose: Improve maturity of development processes
Five Maturity Levels
Level 1: Initial (Ad-hoc and unpredictable processes)
Level 2: Repeatable (Basic project management and tracking)
Level 3: Defined (Organization-wide standard processes)
Level 4: Managed (Quantitative measurement and statistical control)
Level 5: Optimizing (Continuous improvement and innovation)
Role of SQA in Software Maturity
SQA Objectives
Demonstrate confidence in software quality
Prevent defects early
Improve productivity and professionalism
Support process improvement
SQA Growth Across CMM Levels
Level 1: No formal SQA structure (reactive)
Level 2: Basic tracking and regular reviews
Level 3: Organization-wide policies and procedures
Level 4: Quantitative methods and metrics
Level 5: Root cause analysis and prevention
SEI CMM Level 5 in Detail
Characteristics: Continuous improvement and systematic analysis
Key Practices
Defect Prevention: Identify sources and apply corrective actions
Technology Change Management: Evaluate and adopt new tools
Process Change Management: Test and deploy new processes
Benefits: Predictable cost, schedule, and quality
Comparison of ISO 9000 and SEI CMM
ISO 9000: Quality Management System for all industries
SEI CMM: Software process maturity primarily for software
Certification: ISO uses external audits, CMM uses appraisals
Structure: ISO has one level, CMM has five maturity levels
Integration: Can be used together for QMS and process improvement